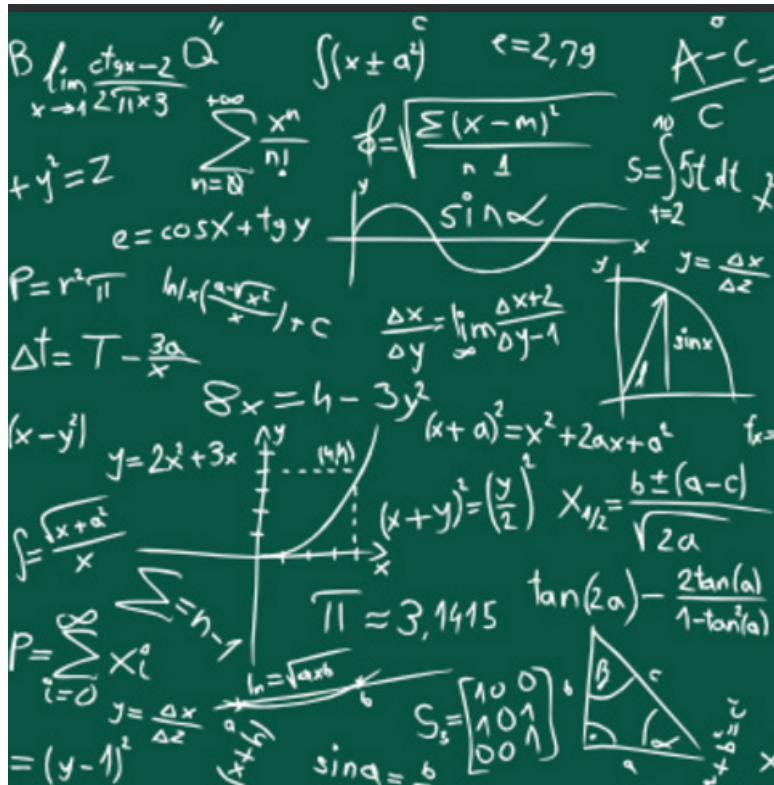


Exercises: summations



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Prove, using the properties of summations, that for every integer $n \geq 1$ we have:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

(sum of the squares of the first n natural numbers, excluding zero).

We start by writing the sum $\sum_{k=0}^n (k+1)^3$ in two different ways:

$$\begin{aligned} 1) \quad \sum_{k=0}^n (k+1)^3 &= \sum_{k=0}^n (k^3 + 3k^2 + 3k + 1) \\ &= \left(\sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + 3 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\ &= \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \frac{3n(n+1)}{2} + n + 1 \end{aligned}$$

$$2) \quad \sum_{k=0}^n (k+1)^3 = \sum_{k=1}^{n+1} k^3 = \sum_{k=1}^n k^3 + (n+1)^3$$

Equating the two expressions and canceling the sum with k^3 , we obtain:

$$3 \sum_{k=1}^n k^2 + \frac{3n(n+1)}{2} + n + 1 = (n+1)^3$$

Isolating the sum with k^2 , we obtain:

$$\begin{aligned} 3 \sum_{k=1}^n k^2 &= (n+1)^3 - \frac{3n(n+1)}{2} - n - 1 \\ &= n^3 + 3n^2 + 3n + 1 - \frac{3n^2 + 3n}{2} - n - 1 \\ &= \frac{2n^3 + 3n^2 + n}{2} \end{aligned}$$

Hence:

$$\sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6}$$

Exercise 2

Prove, using the properties of summations, that for every integer $n \geq 1$ we have:

$$\sum_{k=1}^n k^3 = \frac{n^2 (n+1)^2}{4}$$

(sum of the cubes of the first n natural numbers, excluding zero). That is, we have:

$$\sum_{k=1}^n k^3 = \frac{n^2 (n+1)^2}{4} = \left(\frac{n(n+1)}{2} \right)^2 = \left(\sum_{k=1}^n k \right)^2$$

and hence the sum of the cubes is equal to the square of the sum of the natural numbers.

We start by writing the following sum in two different ways: $\sum_{k=0}^n (k+1)^4$.

$$\begin{aligned} 1) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=0}^n (k^4 + 4k^3 + 6k^2 + 4k + 1) \\ &= \left(\sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\ &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + n(n+1)(2n+1) + 2n(n+1) + n + 1 \\ &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 \\ 2) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=1}^{n+1} k^4 = \sum_{k=1}^n k^4 + (n+1)^4 \end{aligned}$$

Equating the two expressions and canceling the sum with k^4 , we obtain:

$$4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 = (n+1)^4$$

Isolating the sum with k^3 , we obtain:

$$4 \sum_{k=1}^n k^3 = n^4 + 4n^3 + 6n^2 + 4n + 1 - 2n^3 - 5n^2 - 4n - 1 = n^4 + 2n^3 + n^2$$

Hence:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2 (n+1)^2}{4}$$